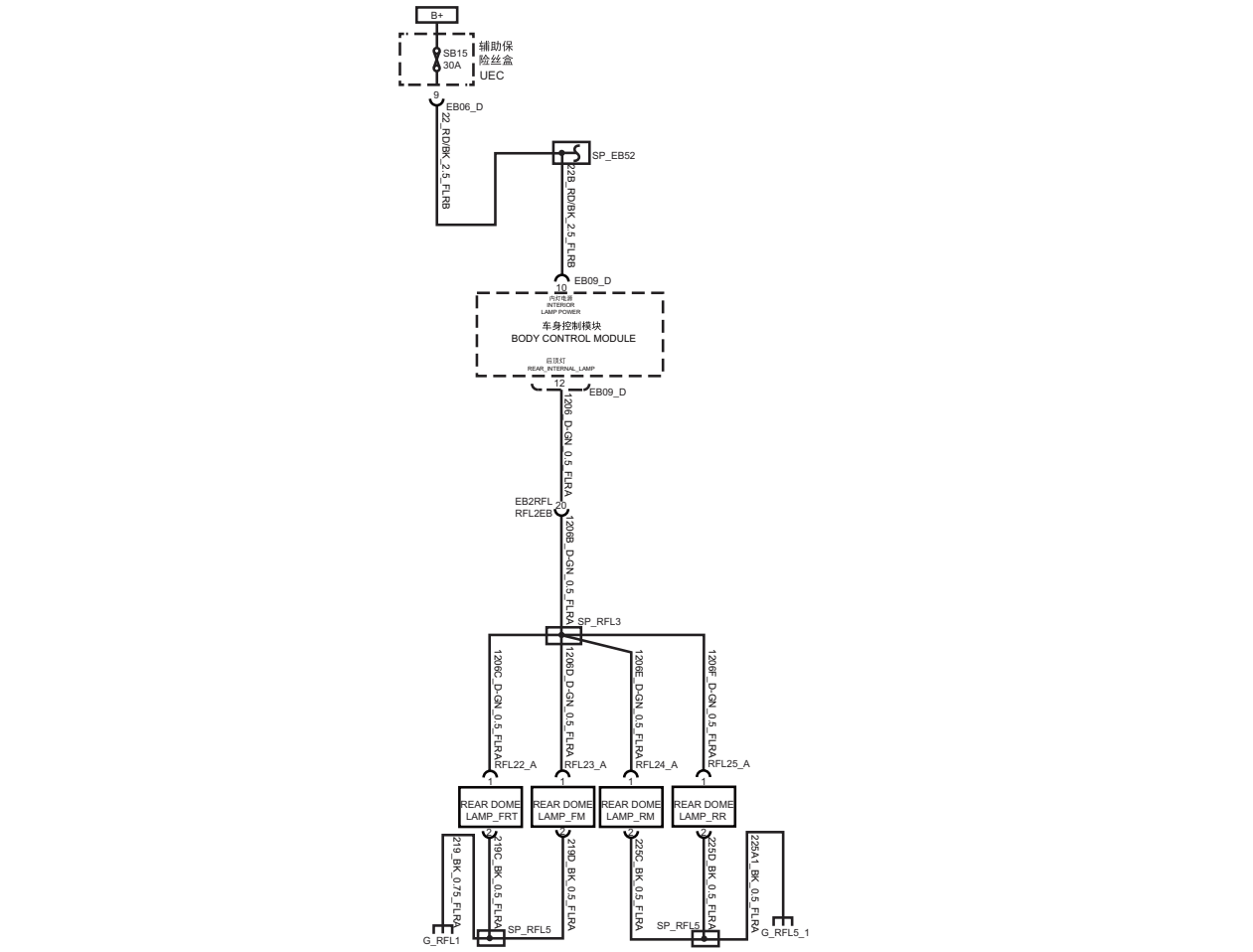
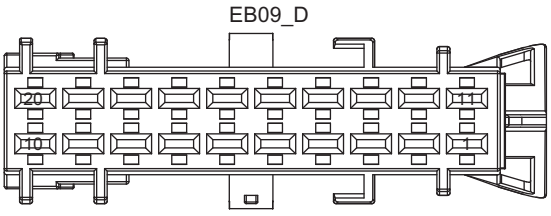
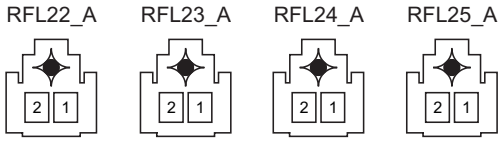
B1112-11、 B1112-12、 B1112-13

Fault diagnosis code
B1112-11：Rear inner light circuit short to ground
B1112-12：Rear inner light circuit short to battery
B1112-13：Rear inner light circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1112-11
Rear inner light circuit short to ground
B1112-12
Rear inner light circuit short to battery
B1112-13
Rear inner light circuit open
To set the effect of a fault code condition
Rear inner light function failure
Description of circuit diagram

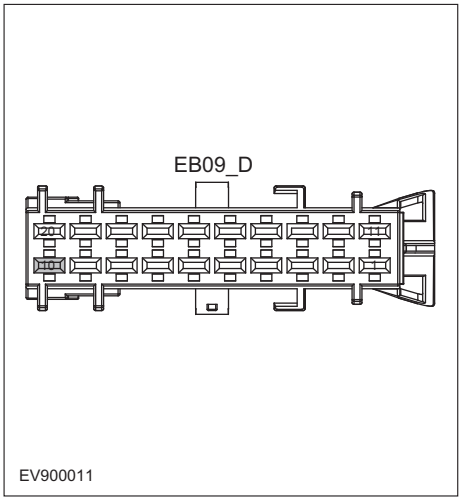
Circuit diagram

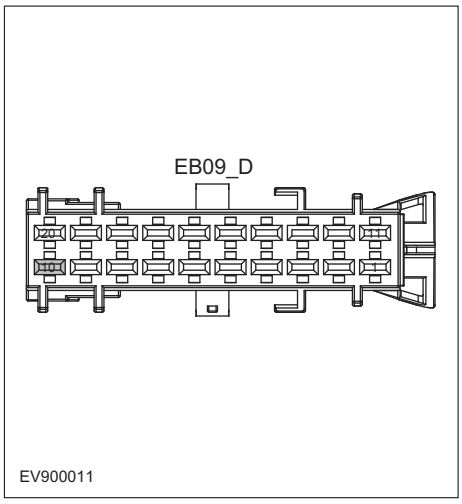
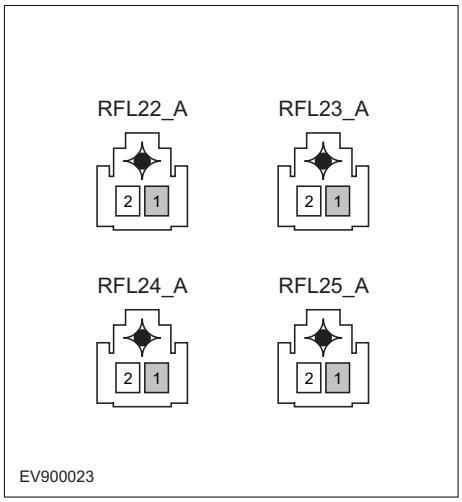


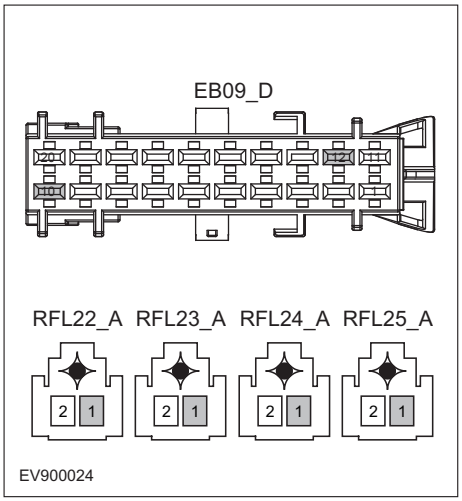


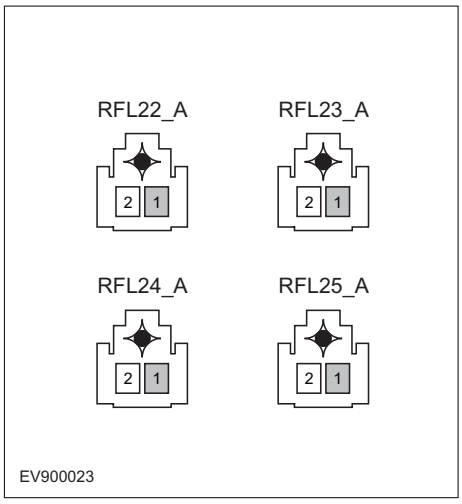


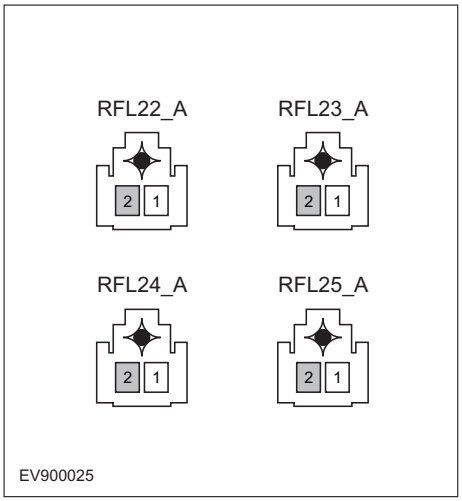
EV900022

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse SB15 for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 Is there any other fault code except for B1112-11/B1112-12/B1112-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the power voltage of inner light of BCM	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D of BCM. 3 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 10 on plug EB09_D and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the power voltage of inner light when the BCM is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 10 on plug EB09_D and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the power voltage of rear overhead light.
	1 Turn the ignition switch to OFF. 2 Connect the harness plug EB09_D of body control module. 3 Disconnect and check the harness plug RFL22_A、RFL23_A、RFL24_A、RFL25_A of rear overhead light. 4 Measure the voltage between the harness terminal of rear overhead light and ground. Measuring circuit: voltage between terminal 1 on plug RFL22_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL23_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL24_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL25_A and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of rear overhead circuit(open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D of BCM. 3 Measure the resistance between the harness terminal of front overhead light and of BCM. Measuring circuit: resistance between terminal 1 on plug RFL22_A and terminal 12 on plug EB09_D. Measuring circuit: resistance between terminal 1 on plug RFL23_A and terminal 12 on plug EB09_D. Measuring circuit: resistance between terminal 1 on plug RFL24_A and terminal 12 on plug EB09_D. Measuring circuit: resistance between terminal 1 on plug RFL25_A and terminal 12 on plug EB09_D. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the circuit of rear overhead circuit(short).
	1 Turn the ignition switch to OFF. 2 Connect the harness plug EB09_D of body control module. 3 Measure the resistance between the harness terminal of rear overhead light and ground. Measuring circuit: resistance between terminal 1 on plug RFL22_A and ground. Measuring circuit: resistance between terminal 1 on plug RFL23_A and ground. Measuring circuit: resistance between terminal 1 on plug RFL24_A and ground. Measuring circuit: resistance between terminal 1 on plug RFL25_A and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of rear overhead light and ground. Measuring circuit: voltage between terminal 1 on plug RFL22_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL23_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL24_A and ground. Measuring circuit: voltage between terminal 1 on plug RFL25_A and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Test whether the grounding circuit of rear overhead light and ground.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of rear overhead light and ground. Measuring circuit: resistance between terminal 2 on plug RFL22_A and ground. Measuring circuit: resistance between terminal 2 on plug RFL23_A and ground. Measuring circuit: resistance between terminal 2 on plug RFL24_A and ground. Measuring circuit: resistance between terminal 2 on plug RFL25_A and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 9	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	